MIL-I-81388A(AS)
2 April 1971
Superseding
MIL-I-81388(AS)
18 July 1966

MILITARY SPECIFICATION

INDICATOR REPEATERS, LIQUID OXYGEN QUANTITY

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers liquid oxygen quantity indicator repeaters having capacities of 5, 10, 20, or 25 liters.

1.2 Classification - Indicator repeaters shall be of the following types (see 6.2):

Type I	- 5 Liter	GMU - 64/A
Type II	- 10 Liter	GMU - 65/A
Type III	- 20 Liter	GMU - 66/A
Type IV	- 25 Liter	GMU - 67/A

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

SPECIFICATIONS

Federal

QQ-P-416 Plating, Cadmium (Electrodeposited)

FSC 1660

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SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-T-152	Treatment, Moisture - and Fungus - Resistant, of Communications, Electronic, And Associated Electrical Equipment
MIL-E-5272	Environmental Testing, Aeronautical And Associated Equipment, General Specification For
MIL-E-5400	Electronic Equipment, Aircraft, General Specification For
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification For
MIL-I-81387	Indicators, Liquid Oxygen Quantity

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

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STANDARDS

Military (Cont'd)

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence For the Selection of
MIL-STD-794	Parts and Equipment, Procedures For Packaging and Packing of
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointer, Dial, Standard Design of Aircraft Instrument
MS33586	Metals, Definition of Dissimilar
MS33639	Cases, Instrument, Clamp-mounted, Aircraft

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government quality control representative. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification - The indicator repeater furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3, 4.3.1, and 6.4). In addition, the retention of the qualification for the indicator repeater on the applicable qualified products list shall be dependent on periodic verification of continued compliance with the requirements of this specification (see 4.3 and 4.3.2).

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3.2 First article - The indicator repeater furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein.

3.3 Selection of specifications and standards - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials - Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.4.1 Metal parts - All metal parts shall be of a corrosion resistance material or treated in a manner to render them adequately resistant to corrosion.

3.4.1.1 Dissimilar metals - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS 33586.

3.4.2 Fungus proof materials - Materials which are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent. Electrical components shall be treated in accordance with MIL-T-152.

3.4.3 Protective treatment- When materials are used in the construction of the indicator repeaters that are subject to deterioration when exposed to environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coating which might crack, chip, or scale during normal service life or under extremes of environmental conditions shall not be used.

3.5 Standard parts - MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the manufacturer's drawings by their part numbers. Parts which are not covered by specifications or military standards, or which are not specifically described herein or on the drawings, shall be completely described on manufacturer's drawings.

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3.6 Design and construction - The indicator repeater shall be designed to operate in conjunction with a master indicator in accordance with MIL-I-81387. The indicator repeater shall consist basically of a synchro receiver which shall be driven by the master indicator.

3.6.1 Case - The indicator repeater case shall be in accordance with MS 33639 and Figure 1. The case shall be made of drawn metal suitable for hermetic sealing and of light construction.

3.6.1.1 Electrical receptacle - The electrical receptacle shall be in accordance with Figure 2 and shall be wired as indicated.

3.6.2 Power failure - The indicator repeater shall provide for power failure indication. A power failure flag, in accordance with Figures 3 and 4 shall appear in the dial cut out when the input power to the indicator is interrupted. The power failure flag shown in Figures 3 and 4 is the preferred location; however, the dial cut out and the dial letters may be relocated provided they are not obscured by the pointer at any pointer position.

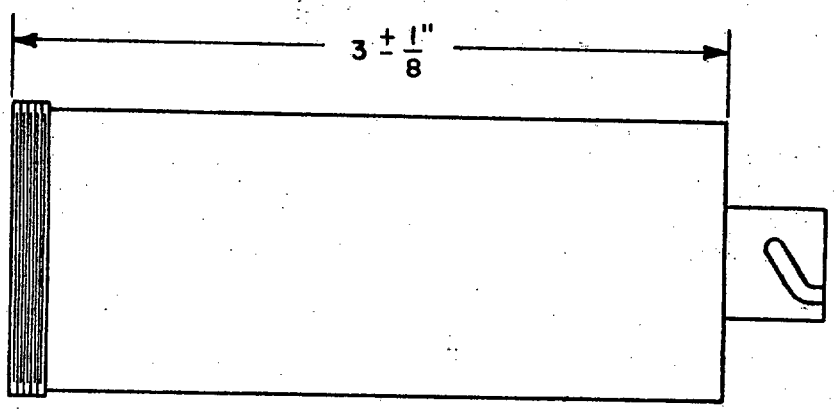
3.6.3 Sealing - The indicator repeater shall be hermetically sealed. The indicator shall be filled with an inert gas composed of at least 10 percent helium and 90 percent nitrogen. The gases shall be of at least 98 percent purity, free of dust particles, and containing not more than 0.006 milligrams of water vapor per liter (dewpoint minus 65°C) at the filling pressure. The absolute pressure of the gas in the case shall be slightly above atmospheric pressure.

3.6.4 Dial - The indicator repeater dial shall be fastened securely to the case or the frame of the mechanism in such a manner that it will not loosen or turn when the indicator repeater is vibrated. If screws are used for attaching the dial, they shall be so located that they will not interfere with any dial marking. The pointer movement shall not be limited by a mechanical stop on the dial.

3.6.4.1 Dial markings - Dial markings shall conform to Figures 3 and 4. The style and proportions of the numerals and letters on the dial shall conform to MS 33558. Numerals shall distinctly indicate the graduation to which they apply.

3.6.4.2 Electrical zero reference mark - The electrical zero position of the synchro receiver shall coincide with the zero quantity indication of the pointer.

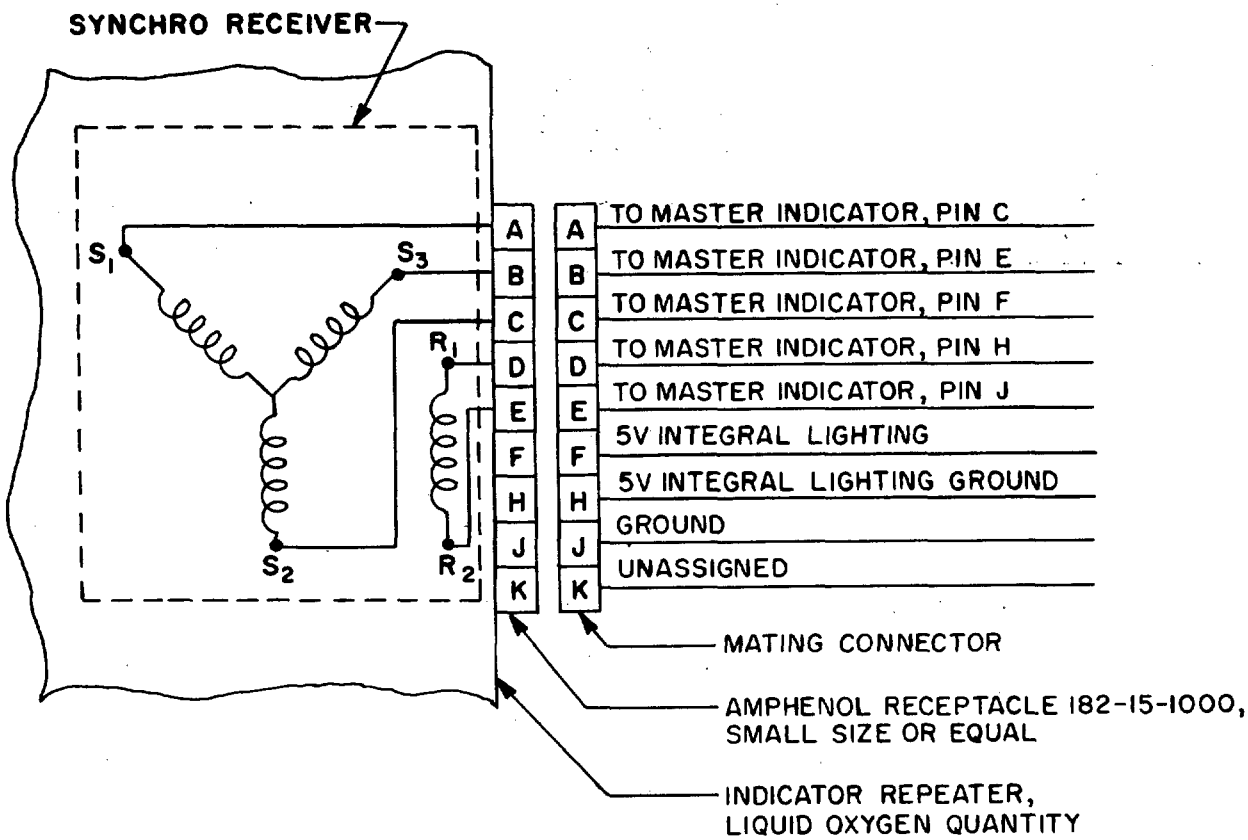
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CASE IN ACCORDANCE WITH MS33639, 2 INCH NOMINAL SIZE

FIGURE 1. INDICATOR CASE

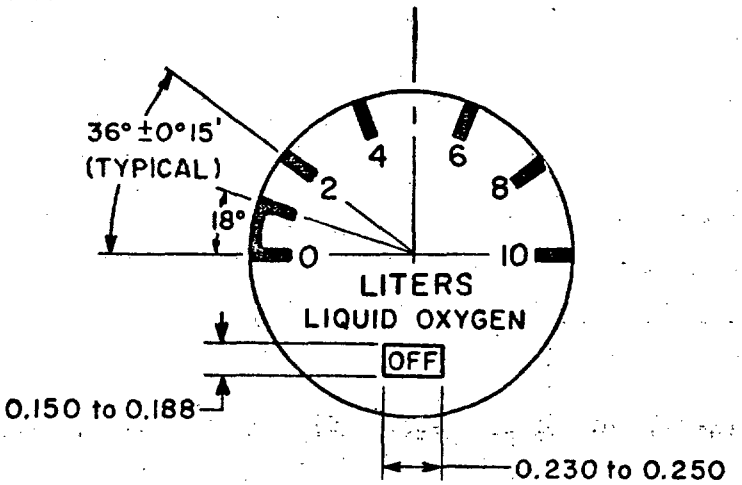
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NOTES: 1. RECEPTACLE INSERT SHALL BE IN NORMAL POSITION.
2. ALTERNATE RECEPTACLES SHALL BE CAPABLE OF MATING WITH AMPHENOL PLUG 165-14-1000, SMALL SIZE.
3. THE SYNCHRO RECEIVER SHALL BE WIRED TO THE RECEPTACLE AS SHOWN ABOVE.

FIGURE 2. ELECTRICAL RECEPTACLE

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10 LITER DIAL



POINTER

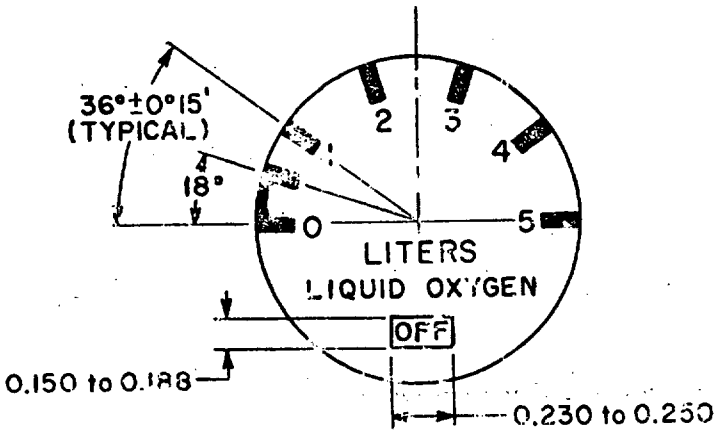
IN ACCORDANCE WITH MS33585-8

DIAL DETAILS (ALL INDICATORS)	HEIGHT	PAINTING
NUMERALS	1/8"	MATTE WHITE FED-STD-595 COLOR NO.37875
GRADUATIONS	3/16" LONG BY .047" WIDE	
"LITERS"	1/8"	
"LIQUID OXYGEN"	3/32"	
"OFF"	3/32"	
POINTER SURFACE		
DIAL BACKGROUND		OPTICAL FLAT BLACK FED-STD-595 COLOR NO.37038
OFF FLAG BACKGROUND		RED FED-STD-595 COLOR NO.31136

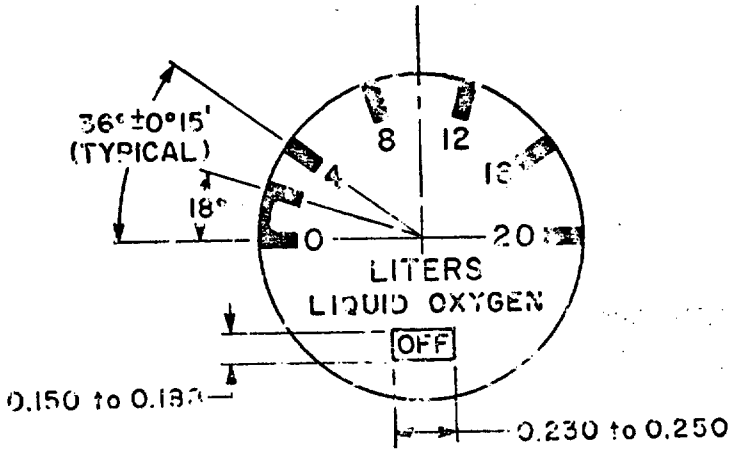
LETTERS AND NUMERALS TO BE IN ACCORDANCE WITH MS33558

FIGURE 3. INDICATOR DIAL AND POINTER

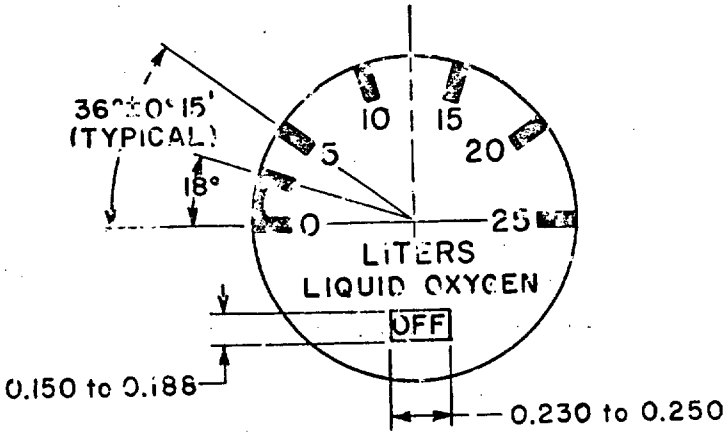
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5 LITER DIAL



20 LITER DIAL



25 LITER DIAL

FIGURE 4. INDICATOR DIALS

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3.6.5 Cover glass - The cover glass of the indicator repeater shall be suitable for hermetic sealing and shall be flat, clear and free from flaws which interfere with normal reading of the indicator repeater. The distance between the front surface of the bezel and the outside surface of the glass shall not exceed 0.062 inch. The distance between the inside surface of the glass and the dial shall not exceed 0.156 inch.

3.6.6 Pointer - The pointer shall extend to, but not into the smallest graduations. The pointer shall be in accordance with MS 33585-8.

3.6.7 Electrical characteristics - The synchro receiver shall receive all operating voltages from the master indicator.

3.6.7.1 Synchro receiver - The synchro receiver shall operate with normal excitation of 115 volts, single phase, 400 cycles, applied to the primary (rotor). The secondary voltage shall be 11.8 ± 0.3 volts when measured between any two secondary leads and the variation between the three secondary voltages shall not exceed 0.2 volt. The power consumption of the synchro shall not exceed 4.60 volt-amperes.

3.6.8 Electronic parts - Electronic parts shall conform to MIL-E-5400.

3.6.9 Integral lighting - A 5 volt AC integral lighting system in accordance with MIL-L-25467 shall be provided.

3.6.10 Screw threads - Screw threads shall conform to MIL-S-7742.

3.7 Performance -

3.7.1 Indicator scale error - The master test instrument reading at each major dial graduation, when tested as specified in 4.7.2, shall be within 1.0 percent of full scale indication with the indicator repeater reading. Full scale indication shall be the difference between the capacitance at zero indication and the capacitance at full indication.

3.7.2 Leakage - The leakage in the hermetic seal of the indicator repeater, when tested as specified in 4.7.3, shall not exceed 0.1 micron cubic foot per hour at a differential pressure of approximately one atmosphere.

3.7.3 Electrical zero - The tip of the indicator repeater pointer, when tested as specified in 4.7.4, shall be within 0.010 inch of the electrical zero reference mark (zero indication) and shall move approximately 30 electrical degrees in the increasing scale direction when pin A is disconnected.

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3.7.4 Dielectric strength - The leakage current between any pin (not wired to the indicator circuit) of the electrical receptacle and the indicator repeater case, when tested as specified in 4.7.5, shall not exceed 1.0 milliampere.

3.7.5 Insulation resistance - The insulation resistance between any pin (except pins F and H) of the electrical receptacle and the indicator repeater case, when tested as specified in 4.7.6, shall be a minimum of 20 megohms.

3.7.6 Power failure- The indicator repeater, when tested as specified in 4.7.7, shall actuate the power failure warning device of the indicator repeater to indicate "OFF" until power is again applied. No part of the power failure flag shall appear in the dial cut out when the input power is supplied to the indicator repeater.

3.7.7 High and low temperature scale error - The indicator repeater, when tested as specified in 4.7.8, shall show no evidence of fogging on the cover glass during the conditioning period. During the conditioning period, the scale error at each major dial graduation shall be within 1.0 percent of full scale indication with the room temperature reference reading. At room temperature the scale error shall be within 0.5 percent of full scale indication with the room temperature reference reading.

3.7.8 Magnetic effect - The deflection of the compass at any of the specified positions, when tested as specified in 4.7.9, shall not exceed five degrees.

3.7.9 Lead length - The indicator repeater, when tested as specified in 4.7.10, shall not change the pointer position more than one degree of arc.

3.7.10 Vibration - The indicator repeater, when tested as specified in 4.7.11, shall not have pointer oscillation or pointer variation exceeding ± 1/32 inch and there shall be no evidence of mechanical or material failure.

3.7.11 Shock test - The indicator repeater, when tested as specified in 4.7.12, shall show no evidence of mechanical or material failure.

3.7.12 Cycling - The indicator repeater, when tested as specified in 4.7.13, shall not exceed the limits specified in the indicator scale error test.

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3.7.13 Thermal shock - The indicator repeater, when tested as specified in 4.7.14, shall show no evidence of moisture penetration or material failure.

3.7.14 Acceleration test - The indicator repeater, when tested as specified in 4.7.15, shall show no evidence of mechanical or material failure.

3.7.15 Radio noise suppression test - The indicator repeater, when tested as specified in 4.7.16, shall not exceed the limits specified in MIL-I-6181.

3.7.16 High and low temperature exposure - The indicator repeater, when tested as specified in 4.7.17, shall maintain a master test instrument reading at each major dial graduation within 0.5 percent of full scale indication with the room temperature reference reading.

3.7.17 Salt spray test - The indicator repeater, when tested as specified in 4.7.18, shall show no evidence of mechanical or material failure.

3.8 Interchangeability - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.9 Weight - The weight of the indicator repeater shall not exceed 0.75 pound.

3.10 Finish -

3.10.1 Aluminum alloy parts - Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625. If such parts are inside the hermetically sealed area, they need not be anodized.

3.10.2 Steel parts - Non-corrosion resistant steel parts shall be cadmium plated where practicable in accordance with QQ-P-416, type II, class 2.

3.11 Identification of product - The indicator repeaters shall be marked for identification in accordance with MIL-STD-130, except that the Federal Stock Number shall be omitted from the nameplate.

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3.11.1 Nameplate - A nameplate shall be securely attached to each separately mounted unit and shall be legibly and permanently marked with the following information:

INDICATOR REPEATER, LIQUID OXYGEN QUANTITY
Type
115 volts, 400 cycle, Single Phase
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's Name or Trademark
U.S.

3.12 Workmanship - The indicator repeaters shall be uniform in quality and shall be free from irregularities, defects, or foreign matter which could adversely affect safety, performance, reliability, or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The examination and testing of the indicator repeaters shall be classified as follows:

(a) Qualification inspection - Qualification inspection consists of examinations and test performed on samples submitted for approval as qualified products (see 3.1, 4.3, 4.3.1 and 6.4).

(1) Retention of qualification - Retention of qualification consists of periodic verification to determine compliance of the qualified product with the requirements of this specification (see 3.1, 4.3 and 4.3.2).

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(b) First article inspection - First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 3.2 and 4.4 through 4.4.1.1).

(c) Quality conformance inspection - Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification.

4.3 Qualification inspection - Qualification inspection shall consist of all the examination and tests of this specification.

4.3.1 Qualification samples - Qualification samples shall consist of two indicator repeaters for which qualification is required. Samples shall be forwarded to the Supply Officer, Naval Air Development Center, Warminster, Pennsylvania 18974, Attention: Director, Aerospace Crew Equipment Department. The samples shall be plainly identified by securely attached durable tags marked with the following information:

Samples for qualification inspection
INDICATOR REPEATER, LIQUID OXYGEN
QUANTITY
Manufacturer's designation or number
Sample submitted by (name) (date) for qualification inspection in accordance with the requirements of MIL-I-81388A(AS) and number under authorization (reference authorizing letter and number; see 6.4).

4.3.2 Retention - The retention of qualification shall consist of periodic verification to determine compliance of the qualified indicator repeater with the requirements of this specification. The time and method of periodic verification shall be specified by the activity responsible for the Qualified Products List and shall be included in the Notice of Qualification letter.

4.4 First article inspection - First article inspection shall consist of the following examinations and tests:

Visual examination
Dimensions
Indicator scale error

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Leakage
Electrical zero
Dielectric strength
Insulation resistance
Power failure
High and low temperature scale error
Lead length

4.4.1 First article samples - Unless otherwise specified, as soon as practicable, after the award of the contract or order, the manufacturer shall submit two indicator repeaters. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these units from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the procuring activity (see 6.2). Approval of the first article samples or the waiving of first article inspection does not preclude the requirements of submitting to the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2).

4.4.1.1 Upon completion of the first article inspection, all the applicable inspection reports and when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the cognizant Government activity. The approved indicator repeaters will be returned to the manufacturer for use in monitoring production.

4.5 Quality conformance inspection - The sampling and inspection levels shall conform to MIL-STD-105. Quality conformance inspection shall consist of the following examinations and tests:

Visual examination
Dimensions
Indicator scale error
Leakage
Electrical zero
Dielectric strength
Insulation resistance
Power failure
Preparation for delivery

4.5.1 Sampling -

4.5.1.1 Inspection lot -

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4.5.1.1.1 Indicator repeater - An inspection lot size shall be expressed in units of indicator repeators, made essentially under the same conditions and from the same materials and components. The sample unit shall be one indicator repeater.

4.5.1.1.2 Preparation for delivery - An inspection lot size shall be expressed in units of one fully prepared shipping container, containing indicator repeaters fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing indicators repeaters fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for tests and examinations of indicator repeaters - The sample size, acceptance criteria, tests, and examinations required for the indicator repeaters shall be as specified in Table I.

4.6 Test conditions -

4.6.1 Temperature and pressure - Unless otherwise specified, tests shall be conducted at local ambient temperature and barometric pressure. The temperature and barometric pressure shall be recorded at the time of inspection. This information shall be available for computation of test data, when required, to normal temperature and pressure (NTP) conditions. NTP conditions are 29.92 inches of mercury and 70°F. Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests.

4.6.2 Tapping - Unless otherwise specified, the indicator repeater may be lightly tapped before a test reading is taken provided the test reading is within 2% of full scale indication to the desired indicator reading.

4.6.3 Test position - Unless otherwise specified, all tests shall be conducted with the indicator repeater in the normal operating position.

4.6.4 Master test instrument - When used, the master test instrument shall be capable of simulating the nominal primary indicator repeater signal within 0.2 percent of full scale indication. The output shall be continuous and calibrated from 0 to 100 percent. Zero percent shall be the equivalent of empty and shall coincide with electrical zero and 100 percent shall be the equivalent of full. In all subsequent tests and measurements specified herein, the tolerance of the measuring equipment shall be added to the specified tolerance for the given test or measurement.

4.6.5 Room temperature reference readings - The room temperature reference readings shall be the master test instrument readings recorded during the indicator scale error test.

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TABLE I

SAMPLE SIZE, ACCEPTANCE CRITERIA, TESTS
AND
EXAMINATIONS OF THE INDICATOR REPEATERS

INSPECTION	PARAGRAPH		SAMPLE SIZE	ACCEPTANCE CRITERIA 1/
	REQUIREMENT	METHOD		
Visual examination (see classification of defects)	3.6 through 3.6.10, 3.8 and 3.11	4.7.1.1	Every indicator repeater for critical defects. Inspection Level II for minor defects.	Reject all units with any critical defects. An acceptable quality level of 2.5 defects per hundred units for minor defects.
Dimensions	3.6.1 and Figures 1, 3 and 4	4.7.1.1.1	Inspection Level S-2	An acceptable quality level of 1.0 defect per 100 units.
Indicator scale error	3.7.1	4.7.2	Every indicator repeater	Reject all defective units
Leakage	3.7.2	4.7.3	Every indicator repeater	Reject all defective units
Electrical zero	3.7.3	4.7.4	Every indicator repeater	Reject all defective units
Dielectric strength	3.7.4	4.7.5	Every indicator repeater	Reject all defective units
Insulation resistance	3.7.5	4.7.6	Every indicator repeater	Reject all defective units
Power failure	3.7.6	4.7.7	Every indicator repeater	Reject all defective units
Preparation for delivery	Section 5	4.7.1.2	Inspection Level S-2	An acceptable quality level of 2.5 defects per 100 units.

1/ The sampling plan acceptance numbers shall apply collectively to all the characteristics within a stated acceptable quality level.

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4.7 Inspection methods -

4.7.1 Visual examination -

4.7.1.1 Indicator repeater - Every indicator repeater shall be examined visually (for critical defects) to determine conformance to this specification. The classification of defects, Table II, shall be used to classify the defects found.

4.7.1.1.1 Dimensions - The indicator repeater shall be checked dimensionally to determine conformance to the dimensions specified herein and on the applicable Military Standard.

TABLE II

CLASSIFICATION OF DEFECTS FOR VISUAL EXAMINATION OF THE INDICATOR REPEATER

Critical	Minor
1. Material imperfections - foreign matter embedded	201. Marking - missing, insufficient, incorrect, illegible, or not permanent.
2. Surface - unclean, rough, misaligned, or containing cracks, nicks, or other flaws,	202. Color not as specified
3. Any component missing, malformed, fractured or otherwise damaged.	
4. Any component loose or otherwise not securely retained.	
5. Incorrect assembling or improper positioning of components.	
6. Any functioning part that works with difficulty.	
7. Faulty workmanship or other irregularities.	

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4.7.1.2 Preparation for delivery - Each of the fully prepared shipping containers, containing indicator repeaters selected as a sample unit from the lot shall be examined to determine that the packaging, packing and marking conform to this specification. The classification of defects, Table III shall be used to classify the defects found.

TABLE III

LIST OF DEFECTS FOR PREPARATION FOR DELIVERY

Item	Defects
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence, or method of application; markings not the same on the interior and exterior containers.
Packaging and Packing materials	Any non-conforming component; any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, container flaps, loose strappings, etc.; bulging or distortion of the containers.
Exterior and interior weight or content	Number per container is more or less than required; gross or net weight exceeds the requirements.

4.7.2 Indicator scale error - The indicator repeater shall be connected to the master test instrument. The indicator repeater pointer shall be set at the center of each major dial graduation by adjusting the master test instrument. The master test instrument reading shall then be recorded. The indicator repeater shall pass the requirements specified in 3.7.1.

4.7.3 Leakage - A mass spectrometer type of helium leak detector shall be used as the method of measuring the leakage. The indicator repeater shall pass the requirements specified in 3.7.2

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4.7.4 Electrical zero - The indicator repeater shall be connected to an electrical zero transformer as shown in Figure 5. The electrical zero transformer shall then be energized and electrical zero imposed on the indicator repeater. Connector pin A (synchro lead S_1) of the indicator repeater shall then be disconnected from the electrical zero transformer. The indicator repeater shall pass the requirements specified in 3.7.3.

4.7.5 Dielectric strength - A potential of 500 volts shall be applied between any pin (not wired to the indicator circuit) of the electrical receptacle and the case of the indicator repeater for a minimum of 5 seconds. The indicator shall pass the requirements specified in 3.7.4. If the indicator repeater case is filled with an inert gas and hermetically sealed, the potential shall be reduced to 300 volts and the same test conducted.

4.7.6 Insulation resistance - The insulation resistance between any pin (except pins F and H) of the electrical receptacle that is wired to the indicator repeater circuit and the case of the indicator repeater shall be measured at 50 volts, direct current. The indicator repeater shall pass the requirements specified in 3.7.5.

4.7.7 Power failure - The indicator shall be connected to the master test instrument. The input power to the indicator repeater shall then be interrupted for approximately 30 seconds and then the power shall be reapplied. The indicator shall pass the requirements specified in 3.7.6.

4.7.8 High and low temperature scale error - The indicator repeater shall be placed in a temperature chamber and subjected to a temperature of 160° ± 5° F for a minimum period of 4 hours. At the end of this period and while still at this temperature, input power shall be applied, and the indicator pointer shall be set at the center of each major dial graduation by means of adjusting the master test instrument. The readings of the master test instrument shall be recorded. The indicator repeater shall then be returned to room temperature and the same scale readings repeated. The indicator repeater shall pass the requirements specified in 3.7.7. The above test procedure shall then be repeated utilizing a temperature of -65° ± 5° F for a minimum period of 4 hours.

4.7.9 Magnetic effect test - The indicator repeater, not operating, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator repeater 5 1/2 inches from, and magnetically east or west of, the center of the compass. Starting directly under compass, the indicator repeater shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees from the initial position. At each of these positions the indicator repeater shall be rotated on its own horizontal axis until it is in its normal upright position.

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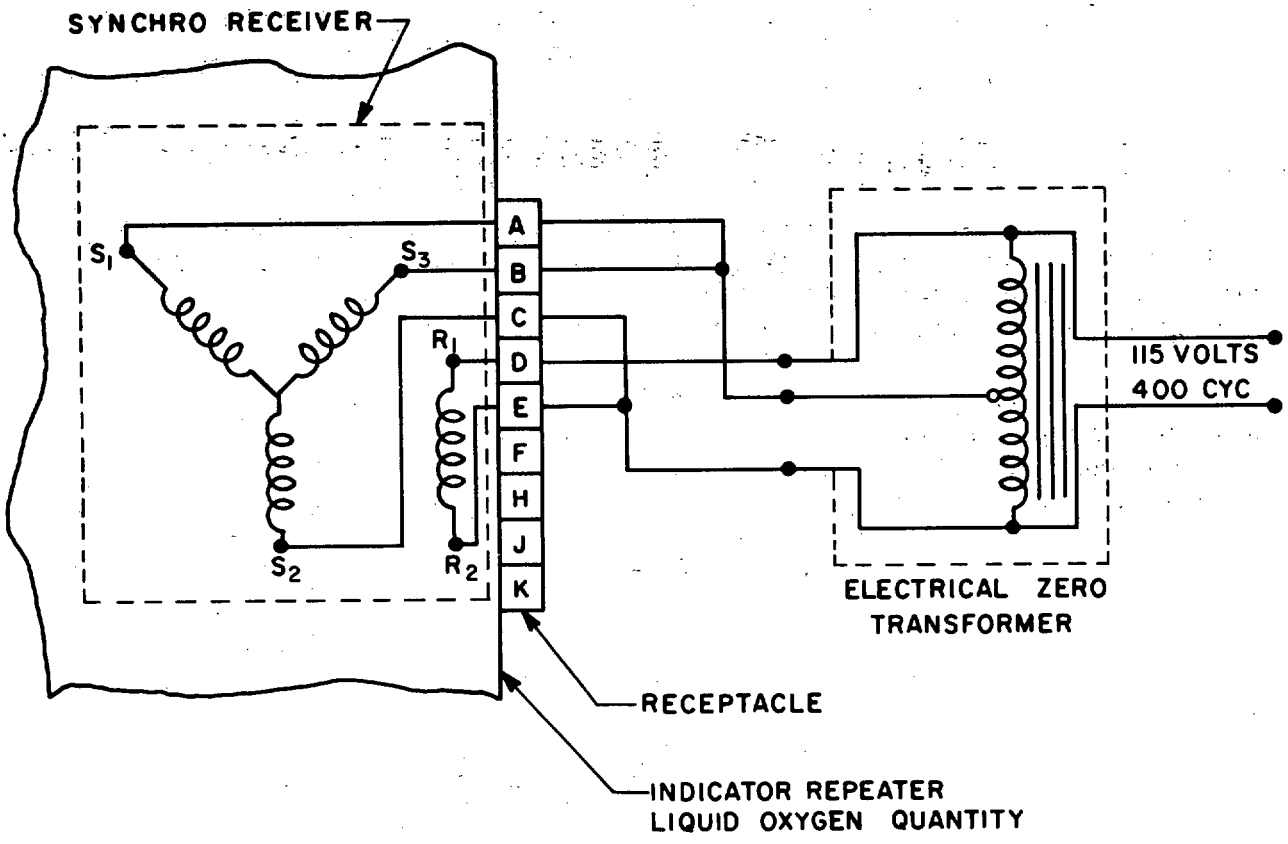


FIGURE 5. ELECTRICAL ZERO TRANSFORMER

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The horizontal magnetic field intensity shall be 0.17 to 0.19 gauss. The indicator repeater shall pass the requirements specified in 3.7.8. The indicator shall then be subjected to the same test procedure with input power supplied.

4.7.10 Lead length - The indicator repeater shall be connected to the master test instrument by a 5 foot length of No. 20 AWG gage wire. The position of the indicator pointer shall be recorded. The wire lead between the indicator and master test instrument shall then be increased to 500 feet and the amount of indicator pointer movement shall be noted. The indicator repeater shall pass the requirements specified in 3.7.9.

4.7.11 Vibration - The indicator repeater shall be subjected to the vibration test specified in MIL-E-5272, Procedure IV and V. Procedure IV shall be conducted with the indicator repeater energized and at three different positions of the indicator pointer: approximately zero point, midscale, and the upperscale end. The indicator shall pass the requirements specified in 3.7.10. The indicator shall then be subjected to and pass the indicator scale error test.

4.7.12 Shock test - The indicator repeater shall be subjected to the shock test (equipment operation) specified in MIL-E-5272, Procedure V. The indicator repeater shall pass the requirements specified in 3.7.11. The indicator repeater shall then be subjected to and pass the following tests:

a. Indicator scale error
b. Sealing
c. Dielectric strength
d. Insulation resistance
e. Power failure

4.7.13 Cycling - The indicator repeater shall be connected to the master test instrument and then subjected to a minimum of 5000 cycles. Each cycle shall consist of continuously varying the quantity indication so that the pointer moves from zero to full and returns to zero within 5 ± 1 minutes. After each 12 cycles, the indicator repeater shall be deenergized electrically for a minimum of 10 minutes. At the conclusion of the cycling, the indicator repeater shall be subjected to the indicator scale error test. The indicator repeater shall pass the requirement specified in 3.7.12.

4.7.14 Thermal shock - The indicator repeater shall be subjected to eight cycles of immersion in tap water maintained at 185 ± 2°F and 40 ± 2°F. Each cycle shall consist of a 30 minute immersion in water maintained at 185 ± 2°F followed by a 30 minute immersion in water maintained at 40 ± 2°F and not more than 5 seconds shall elapse between immersions. The indicator repeater shall pass the requirements specified in 3.7.13. The indicator repeater shall then be subjected to and pass the sealing test.

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4.7.15 Acceleration test - The indicator repeater shall be subjected to the acceleration test specified in MIL-E-5272, Procedure III. The applied acceleration shall be 11 G. The indicator repeater shall pass the requirements specified in 3.7.14. The indicator repeater shall then be subjected to and pass the following test:

a. Indicator scale error
b. Sealing
c. Dielectric strength
d. Insulation resistance
e. Power failure

4.7.16 Radio noise suppression test - The indicator repeater shall be subjected to the radio noise suppression test as specified in MIL-I-6181. The indicator repeater shall pass the requirements specified in 3.7.15.

4.7.17 High and low temperature exposure - The indicator repeater shall be subjected to a temperature of 160° ± 5° F for a minimum of 24 hours. At the end of this period, the indicator repeater shall be returned to room temperature and the indicator pointer set at the center of each major dial graduation by adjusting the master test instrument. The readings of the master test instrument shall be recorded. The indicator repeater shall pass the requirements specified in 3.7.16. The above test procedure shall then be repeated utilizing a temperature of -65° ± 5° F for a minimum period of 24 hours.

4.7.18 Salt spray test - The indicator repeater shall be subjected to the salt spray test specified in MIL-E-5272, Procedure I. The indicator repeater shall pass the requirements specified in 3.7.17.

5. PREPARATION FOR DELIVERY

5.1 Packaging - The indicator repeater shall be packaged in accordance with MIL-STD-794, Level A or C as specified in contract or order (see 6.2). For Level A packaging the method of preservation shall be accordance with MIL-P-116 Method 1A-5, without preservative compound (using metal reusable containers).

5.2 Packing - The indicator repeater shall be packed in accordance with MIL-STD-794, Level A, B, or C as specified in contract or order (see 6.2).

5.3 Marking - The interior and exterior containers shall be marked in accordance with MIL-STD-794.

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5.3.1 Interior container - Each interior container shall be marked with the following information:

INDICATOR REPEATER, LIQUID OXYGEN QUANTITY
Specification MIL-I-81388
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Date of Manufacture
Name of Manufacturer
Name of Contractor (if different from Manufacturer)
"IF IN STORAGE AFTER (DATE)*, THE INDICATOR REPEATER SHALL BE TESTED AND INSPECTED BEFORE USE."

NOTE: *This date shall be 1 year from the date of inspection and shall be inserted by the Inspector at the time of acceptance.

5.3.2 Precautionary marking - The following precautionary markings shall appear conspicuously on two opposite sides of each interior package and shipping container wherever practicable.

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended use - The indicator repeaters covered by this specification are intended to repeat the liquid oxygen quantity indicated by the master indicator.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of indicator repeater required (see 1.2).
(c) Whether first article inspection is required and where the samples should be delivered (see 4.4.1).
(d) Selection of applicable levels of preservation, packaging, and packing.
(e) Applicable methods of cleaning and preservation.
(f) Items of Data required (see 6.3).

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6.3 Data - For the information of Contractors and Contracting Officers, any of the data specified in applicable documents listed in Section 2 of this specification or referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NAVAIR Form 4200/25 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

6.4 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command, Department of the Navy, Washington, D.C. 20360; however, authorization for qualification of products shall be obtained from the Commanding Officer, Naval Air Development Center, Warminster, Pennsylvania 18974, Attention: Director, Aerospace Crew Equipment Department. Prior to submission of the samples for qualification inspection, the manufacturer shall submit a request to the Aerospace Crew Equipment Department indicating a date on which the samples can be forwarded and also request an authorization number to accompany the samples.

6.4.1 When requested, the manufacturer shall submit engineering drawings and inspection reports in accordance with SD-6, Provisions Governing Qualification.

6.5 International standardization - Certain provisions of this specification are the subject of international standardization agreement (ABC 10/2). When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Officers so that appropriate action may be taken respecting the international agreement concerned.

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INDEX

CONTENTS	PARAGRAPH	PAGE
APPLICABLE DOCUMENTS	2	1
CLASSIFICATION	1.2	1
CLASSIFICATION OF DEFECTS		
Preparation for delivery	Table III	18
Visual examination of the indicator repeater	Table II	17
CLASSIFICATION OF INSPECTION	4.2	13
First article inspection	4.2(b)	14
Qualification inspection	4.2(a)	13
Retention of qualification	4.2(a1)	13
Quality conformance inspection	4.2(c)	14
DATA	6.3	24
DESIGN AND CONSTRUCTION	3.6	5
Case	3.6.1	5
Electrical receptacle	3.6.1.1	5
Cover glass	3.6.5	10
Dial	3.6.4	5
Dial markings	3.6.4.1	5
Electrical zero reference mark	3.6.4.2	5
Electrical characteristics	3.6.7	10
Synchro receiver	3.6.7.1	10
Electronic parts	3.6.8	12
Integral lighting	3.6.9	12
Pointer	3.6.6	10
Power failure	3.6.2	5
Screw threads	3.6.10	10
Sealing	3.6.3	5
FIGURES		
Electrical receptacle	Figure 2	7
Electrical zero transformer	Figure 5	21
Indicator case	Figure 1	6
Indicator dial and pointer	Figure 3	6

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Check the source to verify that this is the current version before use.

INDEX

CONTENTS	PARAGRAPH	PAGE
Indicator dials	Figure 4	9
FINISH	3.10	12
Aluminum alloy parts	3.10.1	12
Steel parts	3.10.2	12
FIRST ARTICLE		14
Inspection	4.4	14
Samples	4.4.1	15
Information	4.4.1.1	15
Requirements	3.2	4
IDENTIFICATION OF PRODUCT	3.11	12
Nameplate	3.11.1	13
INSPECTION METHODS	4.7	18
Acceleration test	4.7.15	23
Cycling	4.7.13	22
Dieletric strength	4.7.5	20
Electrical zero	4.7.4	20
High and low temperature exposure	4.7.17	23
High and low temperature scale error	4.7.8	20
Indicator scale error	4.7.2	19
Insulation resistance	4.7.6	20
Lead length	4.7.10	22
Leakage	4.7.3	19
Magnetic effect test	4.7.9	20
Power failure	4.7.7	20
Radio noise suppression test	4.7.16	23
Salt spray test	4.7.18	23
Shock test	4.7.12	22
Thermal shock	4.7.14	22
Vibration	4.7.11	22
Visual examination	4.7.1	18
Indicator repeater	4.7.1.1	18
Dimensions	4.7.1.1.1	18
Preparation for delivery	4.7.1.2	19

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Check the source to verify that this is the current version before use.

INDEX

CONTENTS	PARAGRAPH	PAGE
INTENDED USE	6.1	24
INTERCHANGEABILITY	3.8	12
INTERNATIONAL STANDARDIZATION	6.5	25
MATERIALS	3.4	4
Fungus proof materials	3.4.2	4
Metal parts	3.4.1	4
Dissimilar metals	3.4.1.1	4
Protective treatment	3.4.3	4
ORDERING DATA	6.2	24
PERFORMANCE	3.7	10
Acceleration test	3.7.14	12
Cycling	3.7.12	11
Dielectric strength	3.7.4	11
Electrical zero	3.7.3	10
High and low temperature exposure	3.7.16	12
High and low temperature scale error	3.7.7	11
Indicator scale error	3.7.1	10
Insulation resistance	3.7.5	11
Lead length	3.7.9	11
Leakage	3.7.2	10
Magnetic effect test	3.7.8	11
Power failure	3.7.6	11
Radio noise suppression test	3.7.15	12
Salt spray test	3.7.17	12
Shock test	3.7.11	11
Thermal shock	3.7.13	12
Vibration	3.7.10	11

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:34Z
Check the source to verify that this is the current version before use.

INDEX

CONTENTS	PARAGRAPH	PAGE
PREPARATION FOR DELIVERY	5	23
Marking	5.3	23
Interior container	5.3.1	24
Precautionary marking	5.3.2	24
Packaging	5.1	23
Packing	5.2	23
QUALIFICATION		
Information	6.4	25
Inspection	4.3	14
Retention	4.3.2	14
Samples	4.3.1	14
Requirements	3.1	3
QUALITY CONFORMANCE		
Inspection	4.5	15
Sampling	4.5.1	15
Inspection lot	4.5.1.1	15
Indicator repeater	4.5.1.1.1	16
Preparation for delivery	4.5.1.1.2	16
Sampling for tests and examinations of indicator repeaters	4.5.1.2	16
Sample size, acceptance criteria, test, and examination of the indicator repeaters	Table I	17
RESPONSIBILITY FOR INSPECTION	4.1	13
SCOPE	1.1	1
SELECTION OF SPECIFICATIONS AND STANDARDS	3.3	4
STANDARD PARTS	3.5	4

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Check the source to verify that this is the current version before use.

INDEX

CONTENTS	PARAGRAPH	PAGE
TEST CONDITIONS	4.6	15
Master test instrument	4.6.4	16
Room temperature reference readings	4.6.5	16
Tapping	4.6.2	16
Temperature and pressure	4.6.1	16
Test position	4.6.3	16
WEIGHT	3.9	12
WORKMANSHIP	3.12	13

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SPECIFICATION

MIL-I-81388A(AS) INDICATOR REPEATERS, LIQUID OXYGEN QUANTITY

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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